

Université Paris Dauphine M1 MA
Statistical learning
TD 0 : Introduction to Statistical Learning

Exercise 1: Reminder on Measurability

Let $(\Omega, \mathcal{A}, P)$ be a probability space and let us equip $\mathbb{R}^d$ with the Borel σ -algebra.

Let X be a $\mathbb{R}^d$ -valued random variable and Y a real-valued random variable. Prove that there exists a measurable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $Y = f(X)$ if and only if Y is $\sigma(X)$ -measurable.

Solution:

" $\Rightarrow$ ": Take a measurable function f such that $Y = f(X)$. Let $B \in \mathcal{B}(\mathbb{R})$, we have:

$$Y^{-1}(B) = X^{-1}(f^{-1}(B)).$$

Since f is measurable, the preimage $f^{-1}(B)$ is a Borel set in $\mathbb{R}^d$. Therefore, $X^{-1}(f^{-1}(B))$ is in $\sigma(X)$. Therefore Y is $\sigma(X)$ -measurable.

" $\Leftarrow$ ": If Y is $\sigma(X)$ -measurable, then by the standard result (sometimes called the "technical lemma" or "lemme technique"—see e.g., Proposition 1.1.7 in *Probability: Theory and Examples* by Rick Durrett), any $\sigma(X)$ -measurable function can be written as a countable sum of indicator functions of sets in $\sigma(X)$. This means there exists a measurable function f such that $Y = f(X)$, where f is constructed from the measurable sets in $\sigma(X)$. That is to say, there exists a countable collection of sets $A_i \in \sigma(X)$, $i \in \mathbb{N}$ and real coefficients a_i , $i \in \mathbb{N}$ such that:

$$Y = \sum_{i=1}^{\infty} a_i \mathbf{1}_{A_i}.$$

Here, $\mathbf{1}_{A_i}$ denotes the indicator function of the set A_i , i.e., $\mathbf{1}_{A_i}(\omega) = 1$ if $\omega \in A_i$ and 0 otherwise. Since each $A_i \in \sigma(X)$, there exists a Borel set $B_i \subseteq \mathbb{R}^d$ such that $A_i = X^{-1}(B_i)$, this is true because the preimages of all sets of a sigma algebra by a measurable function forms a sigma algebra, thus $\sigma(X) = \{X^{-1}(B) | B \in \mathcal{B}(\mathbb{R})\}$ (Note that this is not true when X is not measurable). Thus, we can define

$$f(x) = \sum_{i=1}^{\infty} a_i \mathbf{1}_{B_i}(x).$$

To see why f is measurable, note that each $\mathbf{1}_{B_i}$ is measurable as an indicator function of a Borel set, and multiplying by a constant preserves measurability; moreover, the countable sum of measurable functions is measurable because the pointwise limit of measurable functions is measurable. For any $\omega \in \Omega$, we have:

$$f(X(\omega)) = \sum_{i=1}^{\infty} a_i \mathbf{1}_{B_i}(X(\omega)).$$

By definition of $A_i = X^{-1}(B_i)$, we have $X(\omega) \in B_i$ if and only if $\omega \in A_i$, so $\mathbf{1}_{B_i}(X(\omega)) = \mathbf{1}_{A_i}(\omega)$ for each i .

Therefore,

$$\sum_{i=1}^{\infty} a_i \mathbf{1}_{B_i}(X(\omega)) = \sum_{i=1}^{\infty} a_i \mathbf{1}_{A_i}(\omega)$$

Thus, we can write: $Y = f(X)$. □

Exercise 2: Conditional Expectation

Let X and Z be two real-valued random variables. X is assumed to be integrable.

1. Recall the definition of $\mathbb{E}[X | \mathcal{G}]$ when $\mathcal{G}$ is a σ -field.
2. Write the definition of $\mathbb{E}[X | Z]$.
3. Prove that there exists a function $h : \mathbb{R} \rightarrow \mathbb{R}$ measurable such that $\mathbb{E}[X | Z] = h(Z)$ a.s.
4. Write down the exact formula for the function h when Z takes its values in a countable space $E \subseteq \mathbb{R}$.

5. Provide the exact formula for the function h when (X, Z) admits a density $f_{(X,Z)}(x, z)$ with respect to the Lebesgue measure.

Solution:

1,

Let $X \in L^1(\mathcal{F})$. There exists a unique (up to a.s. equality) random variable, denoted $\mathbb{E}[X | \mathcal{G}]$, such that:

- (a) $\mathbb{E}[X | \mathcal{G}]$ is $\mathcal{G}$ -measurable,
- (b) $\mathbb{E}[X | \mathcal{G}]$ is integrable,
- (c) For any bounded, $\mathcal{G}$ -measurable random variable Z ,

$$\mathbb{E}[XZ] = \mathbb{E}[\mathbb{E}[X | \mathcal{G}]Z].$$

2,

$$\mathbb{E}[X|Z] = \mathbb{E}[X|\sigma(Z)].$$

3, By the definition of conditional expectation, and the results of Exercise 1, there exists a measurable function $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\mathbb{E}[X | Z] = h(Z)$ a.s.

4, If Z takes its values in a countable space $E \subseteq \mathbb{R}$, then we can write:

$$h(z) = \sum_{e \in E, \mathbb{P}(Z=e) > 0} \frac{\mathbb{E}[X \mathbf{1}_{\{Z=e\}}]}{\mathbb{P}(Z=e)} \mathbf{1}_{\{e\}}(z)$$

Actually, this formula can be derived this way:

$$\mathbb{E}[X | Z] = \sum_{e \in E} \mathbb{E}[X | Z] \mathbf{1}_{\{Z=e\}}.$$

. Since $\mathbb{E}[X|Z]$ is $\sigma(Z)$ -measurable, $\mathbb{E}[X | Z] \mathbf{1}_{\{Z=e\}}$ can only take values either 0 or $h(e)$ with probability $\mathbb{P}(Z = e)$. By the definition of conditional expectation, we have:

$$\mathbb{E}[X \mathbf{1}_{\{Z=e\}}] = \mathbb{E}[\mathbb{E}[X | Z] \mathbf{1}_{\{Z=e\}}] = h(e) \mathbb{P}(Z = e).$$

Thus

$$h(e) = \frac{\mathbb{E}[X \mathbf{1}_{\{Z=e\}}]}{\mathbb{P}(Z = e)}.$$

when $\mathbb{P}(Z = e) > 0$. By also setting $h(e) = 0$ when $\mathbb{P}(Z = e) = 0$, we can have a well-defined function h that is measurable such that $\mathbb{E}[X \mid Z] = h(Z)$ a.s.

Note that there is another definition of conditional expectation with respect to an event $A \in \mathcal{F}$ as which is defined as expectation of a random variable X with respect to a new probability measure $\mathbb{P}_A$ defined by $\mathbb{P}_A(B) = \mathbb{P}(B \mid A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)} = \int_B \frac{\mathbf{1}_A}{\mathbb{P}(A)} d\mathbb{P}$ for all $B \in \mathcal{F}$ when $\mathbb{P}(A) > 0$:

$$\mathbb{E}[X \mid A] := \mathbb{E}_A[X] = \int X d\mathbb{P}_A = \frac{\int X \mathbf{1}_A d\mathbb{P}}{\mathbb{P}(A)} = \frac{\mathbb{E}[X \mathbf{1}_A]}{\mathbb{P}(A)}.$$

With this notation, we can also write:

$$h(e) = \frac{\mathbb{E}[X \mathbf{1}_{\{Z=e\}}]}{\mathbb{P}(Z = e)} = \mathbb{E}[X \mid Z = e].$$

5, If (X, Z) admits a density $f_{(X,Z)}(x, z)$ with respect to the Lebesgue measure, then we can write:

$$h(z) = \int_{\mathbb{R}} x \frac{f_{(X,Z)}(x, z)}{f_Z(z)} \mathbf{1}_{f_Z(z) > 0} dx.$$

This can be derived the following way: Take $B \in \mathcal{B}(\mathbb{R})$, then we have:

$$\begin{aligned} \int_{\mathbb{R}} h(z) \mathbf{1}_B(z) f_Z(z) dz &= \mathbb{E}[h(Z) \mathbf{1}_{\{Z \in B\}}] \\ &= \mathbb{E}[X \mathbf{1}_{\{Z \in B\}}] \\ &= \int_{\mathbb{R}^2} x \mathbf{1}_B(z) f_{(X,Z)}(x, z) dx dz \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} x f_{(X,Z)}(x, z) dx \right) \mathbf{1}_B(z) dz. \end{aligned}$$

By arbitrary choice of B , for almost all $z \in \mathbb{R}$, we have

$$\int_{\mathbb{R}} x f_{(X,Z)}(x, z) dx = h(z) f_Z(z).$$

If we have $f_Z(z) = 0$, i.e., $\int_{\mathbb{R}} f_{(X,Z)}(x, z) dx = 0$, we have

$$f_{(X,Z)}(x, z) = 0$$

for almost all x in $\mathbb{R}$, thus

$$xf_{(X,Z)}(x, z)\mathbf{1}_{f_Z(z)>0} = 0$$

for almost all x in $\mathbb{R}$, thus its integral will become zero. Therefore, if $f_Z(z) = 0$, we can simply take $h(z) = 0$. If $f_Z(z) > 0$, we can divide both sides by $f_Z(z)$ to conclude.

□

Exercise 3: Conditional Expectation - Examples

1. Let X and Y be two independent, identically distributed and integrable random variables. Prove that $\mathbb{E}[X \mid X + Y] = \frac{X+Y}{2}$ a.s.
2. Let X_1 and X_2 be two independent Poisson variables with parameters λ_1 and λ_2 .
 - (a) What is the distribution of X_1 given $X_1 + X_2$?
 - (b) Calculate $\mathbb{E}[X_1 \mid X_1 + X_2]$.
3. Let $X_1, X_2, X_3 \sim_{i.i.d.} \mathcal{E}(\alpha)$. Give the law of the pair $(X_2 - X_1, X_3 - X_1)$ given X_1 .

Solution:

1. First, note that $\frac{X+Y}{2}$ is $\sigma(X + Y)$ -measurable. Take any bounded, $\sigma(X + Y)$ -measurable function g . Since X and Y are independent and identically distributed, the pair (X, Y) has the same distribution as (Y, X) . Therefore, $(X, X + Y)$ has the same distribution as $(Y, X + Y)$. This implies $\mathbb{E}[g(X + Y)X] = \mathbb{E}[g(Y + X)Y]$. So, $\mathbb{E}[g(X + Y)X] = \mathbb{E}[g(X + Y)Y]$. Summing them up, we get $2\mathbb{E}[g(X + Y)X] = \mathbb{E}[g(X + Y)(X + Y)]$. By the definition of conditional expectation, this means

$$\mathbb{E}[X \mid X + Y] = \mathbb{E}[Y \mid X + Y] = \frac{X + Y}{2} \quad \text{a.s.}$$

2. (a) $X_1 + X_2$ takes values in $\mathbb{N}$, and is a Poisson variable with parameter $\lambda_1 + \lambda_2$. Fix $n \in \mathbb{N}$, for any $k \in \llbracket 0, n \rrbracket$, we have

$$\begin{aligned} P(X_1 = k \mid X_1 + X_2 = n) &= \frac{P(X_1 = k, X_2 = n - k)}{P(X_1 + X_2 = n)} \\ &= \frac{\frac{(\lambda_1)^k e^{-\lambda_1}}{k!} \cdot \frac{(\lambda_2)^{n-k} e^{-\lambda_2}}{(n-k)!}}{\frac{(\lambda_1 + \lambda_2)^n e^{-(\lambda_1 + \lambda_2)}}{n!}} \\ &= \frac{n!}{k!(n-k)!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}. \end{aligned}$$

This is the probability mass function of a binomial distribution with parameters n and $p = \frac{\lambda_1}{\lambda_1 + \lambda_2}$. So, $X_1 \mid (X_1 + X_2 = n) \sim \mathcal{B}(n, p)$.

- (b) From the result above, the conditional expectation is the expectation of the corresponding binomial distribution:

$$\mathbb{E}[X_1 \mid X_1 + X_2 = n] = n \cdot p = n \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Thus, as a random variable,

$$\mathbb{E}[X_1 \mid X_1 + X_2] = (X_1 + X_2) \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

3. Let $x > 0$, and $(Y_2, Y_3) := (X_2 - X_1, X_3 - X_1)$. We want to find the conditional distribution of (Y_2, Y_3) given $X_1 = x$. Take an arbitrary bounded continuous function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$. We have:

$$\begin{aligned} \mathbb{E}[g(Y_2, Y_3) \mid X_1 = x] &= \mathbb{E}[g(X_2 - x, X_3 - x) \mid X_1 = x] \\ &= \mathbb{E}[g(X_2 - x, X_3 - x)] \\ &= \int_{\mathbb{R}^2} g(y_2, y_3) \alpha^2 e^{-\alpha(y_2 + x + y_3 + x)} \mathbf{1}_{y_2 > -x, y_3 > -x} dy_2 dy_3, \end{aligned}$$

where the second last equality follows from the fact that X_1, X_2 and X_3 are independent. By arbitrary choice of g , we have:

$$f_{(X_2 - X_1, X_3 - X_1) \mid X_1 = x}(y_2, y_3) = \alpha^2 e^{-\alpha(y_2 + x + y_3 + x)} \mathbf{1}_{y_2 > -x, y_3 > -x}.$$

This means that the conditional distribution of $(X_2 - X_1, X_3 - X_1)$ given $X_1 = x$ is the same as the joint distribution of two independent exponential random variables with parameter α , shifted by x .

□

Exercise 4: Conditional independence

Let X and Y be two random variables in a measure space $(E, \mathcal{E})$ and let $\mathcal{G}$ be a sigma-algebra. X and Y are independent given $\mathcal{G}$ if for all measurable functions f and g from E to $\mathbb{R}^+$,

$$\mathbb{E}[f(X)g(Y) \mid \mathcal{G}] = \mathbb{E}[f(X) \mid \mathcal{G}]\mathbb{E}[g(Y) \mid \mathcal{G}]$$

1. What does this mean if $\mathcal{G} = \{\emptyset, \Omega\}$?
2. Prove that the above definition is equivalent to: for any positive $\mathcal{G}$ -measurable random variable Z , and for any measurable functions f and g from $\mathbb{R}$ to $\mathbb{R}^+$,

$$\mathbb{E}[f(X)g(Y)Z] = \mathbb{E}[f(X)Z\mathbb{E}[g(Y) \mid \mathcal{G}]].$$

3. Prove that the above definition is also equivalent to: for any measurable function g from $\mathbb{R}$ to $\mathbb{R}^+$,

$$\mathbb{E}[g(Y) \mid \sigma(\mathcal{G}, X)] = \mathbb{E}[g(Y) \mid \mathcal{G}].$$

Solution:

1, If $\mathcal{G} = \{\emptyset, \Omega\}$, then the conditional expectation $\mathbb{E}[f(X)g(Y) \mid \mathcal{G}]$ is simply $\mathbb{E}[f(X)g(Y)]$, which is a constant. To verify this, we can take a random variable $Z = 1$, then we have $\mathcal{G} = \sigma(Z)$. Thus any conditional expectation with respect to $\mathcal{G}$ is of the form $h(Z)$ for a measurable function h , i.e., constant a.s.. By taking expectation again, we have that the conditional expectation with respect to $\mathcal{G}$ is equal to the unconditional expectation.

The independence condition simplifies to:

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)],$$

which is one of the definition of independence of X and Y .

2, Suppose the above definition holds. Let Z be a positive $\mathcal{G}$ -measurable random variable. Then, for any measurable functions f and g , we have:

$$\mathbb{E}[f(X)g(Y)Z] = \mathbb{E}[\mathbb{E}[f(X)g(Y) \mid \mathcal{G}]Z] = \mathbb{E}[\mathbb{E}[f(X) \mid \mathcal{G}]\mathbb{E}[g(Y) \mid \mathcal{G}]Z].$$

Since $\mathbb{E}[g(Y) \mid \mathcal{G}]Z$ is also a positive $\mathcal{G}$ -measurable random variable, we can write:

$$\mathbb{E}[f(X)g(Y)Z] = \mathbb{E}[f(X)Z\mathbb{E}[g(Y) \mid \mathcal{G}]].$$

Conversely, suppose the second condition holds. Since $\mathbb{E}[g(Y) \mid \mathcal{G}]Z$ is also a positive $\mathcal{G}$ -measurable random variable, we can write:

$$\mathbb{E}[f(X)\mathbb{E}[g(Y) \mid \mathcal{G}]Z] = \mathbb{E}[\mathbb{E}[f(X) \mid \mathcal{G}]\mathbb{E}[g(Y) \mid \mathcal{G}]Z] = \mathbb{E}[f(X)g(Y)Z].$$

Thus the first condition holds.

3, Suppose the above definition holds. Let

$$\mathcal{C} = \{A \in \sigma(\mathcal{G}, X) \mid \mathbb{E}[\mathbf{1}_A g(Y)] = \mathbb{E}[\mathbf{1}_A \mathbb{E}[g(Y) \mid \mathcal{G}]]\}.$$

On the one hand, we to show that $\mathcal{C}$ is a monotone class, or λ -system, i.e., it is closed by difference and monotone countable union, and contains the Ω . Closedness by difference is from linearity of expectation. Closedness by monotone countable union is from the monotone convergence theorem. It obviously contains Ω .

On the other hand, we want to show that $\mathcal{P} := \{A \cup B \mid A \in \sigma(X), B \in \mathcal{G}\}$ is a π -system, and is a subset of $\mathcal{C}$. $\mathcal{P}$ is a π -system because it is closed by intersection, and contains the empty set. Take $A \in \sigma(X)$ and $B \in \mathcal{G}$. $\mathbf{1}_A$ can be written in the form of $f(X)$ for some positive measurable function f , and $\mathbf{1}_B$ is $\mathcal{G}$ -measurable. Thus, we have:

$$\mathbb{E}[\mathbf{1}_A g(Y) \mathbf{1}_B] = \mathbb{E}[f(X)g(Y) \mathbf{1}_B] = \mathbb{E}[f(X)\mathbb{E}[g(Y) \mid \mathcal{G}] \mathbf{1}_B].$$

Therefore, we have:

$$\mathbb{E}[\mathbf{1}_{A \cap B} g(Y)] = \mathbb{E}[\mathbf{1}_A g(Y) \mathbf{1}_B] = \mathbb{E}[\mathbf{1}_A \mathbb{E}[g(Y) \mid \mathcal{G}] \mathbf{1}_B] = \mathbb{E}[\mathbf{1}_{A \cap B} \mathbb{E}[g(Y) \mid \mathcal{G}]].$$

,i.e., $A \cap B \in \mathcal{C}$. By The monotone class theorem, we have $\mathcal{C}$ contains $\sigma(\mathcal{P})$, i.e., $\sigma(\mathcal{G}, X) \subseteq \mathcal{C}$. By the definition of conditional expectation, we have:

$$\mathbb{E}[g(Y) \mid \sigma(\mathcal{G}, X)] = \mathbb{E}[g(Y) \mid \mathcal{G}].$$

Conversely, suppose the third condition holds. Let $B \in \mathcal{G}$, we have $\mathbf{1}_B f(X)$ is $\sigma(\mathcal{G}, X)$ -measurable, and thus we can write:

$$\mathbb{E}[\mathbf{1}_B f(X)g(Y)] = \mathbb{E}[\mathbf{1}_B f(X)\mathbb{E}[g(Y) \mid \sigma(\mathcal{G}, X)]] = \mathbb{E}[\mathbf{1}_B f(X)\mathbb{E}[g(Y) \mid \mathcal{G}]].$$

Since $\mathbf{1}_B f(X)\mathbb{E}[g(Y) \mid \mathcal{G}]$ is $\mathcal{G}$ -measurable, we can write:

$$\mathbb{E}[\mathbf{1}_B f(X)\mathbb{E}[g(Y) \mid \mathcal{G}]] = \mathbb{E}[\mathbf{1}_B \mathbb{E}[g(Y) \mid \mathcal{G}]\mathbb{E}[f(X) \mid \mathcal{G}]].$$

Thus we have:

$$\mathbb{E}[\mathbf{1}_B f(X)g(Y)] = \mathbb{E}[\mathbf{1}_B \mathbb{E}[g(Y) \mid \mathcal{G}] \mathbb{E}[f(X) \mid \mathcal{G}]].$$

By the definition of conditional expectation, we have:

$$\mathbb{E}[f(X)g(Y) \mid \mathcal{G}] = \mathbb{E}[f(X) \mid \mathcal{G}] \mathbb{E}[g(Y) \mid \mathcal{G}].$$

Therefore, the first condition holds.

□